SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA14 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Mode: In Class **Hours : 60 mts**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Keerthana. R (192521446) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q. No	Problem Understanding (20)	Method Selection (20)	Solution Process (25)	Accuracy of Calculation (20)	Interpretation of Results (15)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 20	/ 25	/ 20	/ 15	/ 100

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO - 1 Assessment tool - 1

CSA1410 - Compiler Design

Name : Keerthana.G

Reg no : 192581446

Dept : IT

90/100

03/10/24

Analytical problem solving

1. Trace the processing of input expression $p = (a * b) + (b * c) * 2$ through all phases

a) Lexical Analysis - Tokens

p - Identifier

= - Assignment operator

(- left parenthesis

a - Identifier

* - Multiplication operator

b - identifier

) - right parenthesis

+ - Addition operator

(- left parenthesis

b - Identifier

*

(- Identifier

) - right parenthesis

* - multiplication operator

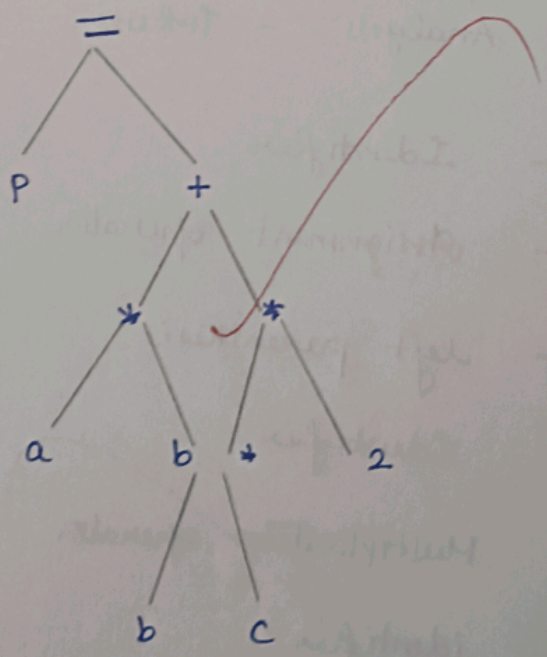
2 - Number

; - Terminator

id = (id + id) + (id + id) * 2

b) Syntax Tree

P = (a + b) + (b + c) * 2



c) Intermediate code.

t_1 = a * b

t_2 = b * c

$t_3 = t_2 * 2$

$t_4 = t_1 + t_3$

$p = t_4$

d) code optimization

$t_1 = a * b$

$t_2 = b * c$

$t_3 = t_2 << 1$

$t_4 = t_1 + t_3$

$p = t_4$

e) Significance of Final target code.

```
MOV  R1, a
MUL  R1, b
MOV  R2, b
MUL  R2, c
SHL  R2, 1
ADD  R1, R2
MOV  P, R1
```

2. For each of the following regular expressions, construct the language. Write any five valid string for each regular expression

a) aa

$L = \{ \epsilon, aa, aaaa, aaaaaa \}$

five valid strings:

aa
aaaa
aaaaaa
aaaaaaaa
aaaaaaaaaa

b) $(b^*)^* abb$

$(b^*)^* = b^*$

$b^* abb$

five valid strings:

abb
babb
bbabb
bbbabb
bbbbabb

c) ~~$((a^*)^*)^*$~~ $(a^*b^*)^* aba (a^* + b^*)^*$

five valid string

aba
aaba

$L = \{ \varepsilon, ab, abab, ababab \ldots \}$

five valid strings

ab
ab ab
ab ab ab
ab ab ab ab
ab ab ab ab ab

e) $(aba)^* + (a^+ b^+)^+$

five valid strings

a
b
a b
b a
a a b

---

3. find the number of tokens in the following c statement

i) Number of tokens.

Main , (,) , { , int , a , = , 10 , ;
char , b , = , "abc" , ; , int , c , = , 30 , ;
char , d , = , "xyz" , ; , int , m ,
= , 40.5 , ; , }

The total Number of tokens are 30

ii) identify the lexical error.

```
void main ()
{
    int x=10, y=20;
    char * a;
    a = &x;
    x = 1 x ab;
}.
```

The lexical error : 1 x ab

It is Invalid because identifier cannot start with digit

4. find the regular Expression corresponding to each the following subset of {0,1}:

$\Sigma = \{0, 1\}$

i) strings having an even number of 0's

$1^* (01^* 01^*)^*$

ii) Strings containing at least one

$0^* 1 (0+1)^*$

iii) The language of all strings in which both the numbers of 0's

and the number of 1's are odd.

$$(00+11)^* (01+10)(00+11)^* [(01+10)(00+11)^* (01+10)(00+11)^*]^*$$

iv) the language of all strings that start with 1 and end with 0

$$1(0+1)^*0$$

Example : 10, 100, 1010

v) the language of all strings that do not contain consecutive 1's

$$(0+10)^* (\epsilon+1)$$

---

5. Consider a lexical Analyzer for simplified programming language.

a) How many tokens are there.

int, var123, =, 12345, ;, float, var456, =, 123.45, ;, if, var123, =!=, var456, {, return, ;

The total Number of tokens are 20.

b) categorize the tokens

1) keywords.

int, float, if, return

4 - keywords

2) operators

=, =, ==

3 - operators

3) separators

;, ;, (,), {, ;, }

7 - operators

4) Identifiers

var123, var456, var123, var456

4 - Identifiers

5) Integer literals

12345

1 - Integer literals

6) floating - point literals.

123.45

1 - floating - point literals

Also totally 20 tokens.

c) FSM state transition.

For each token FSM is based on ch

Token	No. of transition.
int	3
var123	6
=	1
12345	5
;	1
float	5
var456	6
=	1
123.45	6
;	1
if	2

var 123	6
= =	2
var456	6
)	1
{	1
return	6
;	1
}	1.

Totally 62 transitions.